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Synthetic End Item Data Package (Debug Reference)

This document intentionally blends nominal, off-nominal, and incomplete data so the scanner can be exercised against a broad set of scenarios (duplicate terms, missing units, multi-value tables, etc.). Only synthetic values are used.

Document Profile

Field	Value
Program	Hyperion Dragonfly Propulsion Demo
Vehicle	SV3 (Block 2)
Serial / Component	SN42-AX
Build Revision	Rev E (04)
Date Compiled	2025-01-17
Compiler	QA Test Harness
Source Bundle	/Data Packages/Hyperion_SV3_SN42_RevE

1. Functional Acceptance Snapshot

Term	Description	Requirement	Measured Value	Units	Page	Data Quality	Notes
ignition_delay	Time between arm and ignition	≤ 0.85	0.62	s	5	PASS	sample nominal case

Term	Description	Requirement	Measured Value	Units	Page	Data Quality	Notes
isp	Steady-state specific impulse	43 ± 3	39.4	sec	6	LOW	flagged low but inside tolerance window
chamber_pressure	Mean P _{cc}	185 ± 10	201	psia	6	WARN	value above requirement, watch for drift
burn_duration	Hot-fire duration	42–48	0 (not testable)	s	6	NOT_RUN	use to test textual statuses
resistance	Injector harness resistance	4.8–5.3	6.1	Ω	8	FAIL	triggers out-of-range numeric failure
valve_alignment	Valve alignment photo	Visual match	Image #12	n/a	9	PASS	non-numeric text extraction
leak_check	Helium leak rate	≤ 1.0e-5	Not Found	atm·cc/s	10	MISSING	ensures missing term handling

2. Performance KPI Table

KPI	Min	Target	Max	Value	Source	Confidence
thrust_nominal	4.8	5.1	5.5	5.32	Table 3	0.97
thrust_peak	–	6.2	6.8	6.95	Table 3	0.91
power_draw_idle	75	80	95	104	Table 4	0.77
power_draw_peak	–	140	155	131	Table 4	0.85
thermal_margin	8	12	–	6	Fig. 5	0.63

Use this table to confirm multi-column parsing and comparisons against asymmetric spec bands (no min or max defined).

3. Environmental Endurance Matrix

Test	Temp (°C)	Duration (hr)	Requirement Met?	Deviation	Log Ref
Thermal Soak Low	-45	2.0	YES	±0.5	A-112
Thermal Soak High	+95	1.5	NO	Chamber tripped at +88°C	A-118
Thermal Ramp Cycling	-35↔+85	6 cycles	YES	–	A-126
Random Vib (X)	14.1 grms	7 min	YES	–	V-031
Random Vib (Z)	12.0 grms	7 min	CONDITIONAL	Sensor SAT	V-035
Sine Sweep	5–500 Hz	0.5 g	Not Recorded	–	V-041

Mix of numeric, range (arrow), textual, and missing entries stresses parser assumptions.

4. Calibration & Trim Table

Channel	Pre Trim	Post Trim	Allowed Delta	Status	Comment
TC-01	1.0034	0.9991	±0.0100	PASS	Temperatures within tolerance
TC-02	0.9989	1.0142	±0.0100	FAIL	Out-of-family high
PT-03	1.0000	1.0000	±0.0020	PASS	Perfect match
PT-07	0.9977	0.9974	±0.0020	INVESTIGATE	Delta close to limit
LVDT-2	1.0040	–	±0.0050	INCOMPLETE	Missing post-trim data

5. Anomaly & Waiver Log

ID	Severity	Term	Description	Containment	Status
DR-2025-014	Major	chamber_pressure	Persistent +8% over spec during burn #2	Added relief orifice, retest pending	OPEN
NCR-25-008	Minor	resistance	Harness swapped due to 6.1 Ω reading	Re-test verified 5.05 Ω	CLOSED
WA-25-002	Info	leak_check	Leak test deferred pending new fixture	Approved by Chief QA	APPROVED

6. Attachment Index (Synthetic)

Attachment	File Name	Pages	Notes
Acceptance Test Report	ATR_Hyperion_SV3_SN42.pdf	124	Contains Sections 3 & 4 tables
Thermal Test Log	THERM_SV3_SN42_revC.pdf	56	Page numbers offset by +2 in footer
Vib Test Photos	VIB_IMG_SN42.zip	24 images	Use to test non-PDF assets
Calibration Data	CAL_SN42_revE.xlsx	4 sheets	Multi-tab workbook

Usage Hints

- 1. Place this Markdown file alongside PDFs to validate mixed content detection.
- 2. Convert to PDF (e.g., via pandoc) if you want to test page-scoped extraction.
- 3. Adjust term lists so at least one expected value lives in every section above.